Unit 05: Profit and Loss

CONTENTS

Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We have learnt how to calculate Cost Price, Selling Price and Profit or Gain.
- We have learnt how to calculate successive discount and marked price

Introduction

Nowadays, transactions have become a common feature of life. When a person deals in the purchase and sale of any item, he either gains or loses some amount generally. The aim of any business is to earn profit.

Profit (P)

The amount gained by selling a product with more than its cost price.

Loss (L)

The amount that you lose after selling a product with less than its cost price.

Cost Price – This is the rate at which the commodity is bought. The amount paid for a product or commodity to purchase it is called a cost price. It can be divided into two parts:

Fixed Cost

Variable Cost

Selling Price – This is the rate at which the commodity is sold.

Profit – When the Cost Price of the commodity is less than the Selling Price.

Profit = Selling Price – Cost Price

Loss – When the Cost Price of the commodity is greater than the Selling Price.

Loss = Cost Price – Selling Price

If there is a PROFIT of x%, the calculating figures would be 100 and (100 + x).

If there is a LOSS of x%, the calculating figures would be 100 and (100 - x).

Calculating figures be Cost Price and Selling Price respectively.

The commonlyused terms in dealing with questions involving sale and purchase are:

Cost Price: The cost price of an article is the price at which an article has been purchased. It is abbreviated as C.P.

Selling Price: The selling price of an article is the price at which an article has been sold. It is abbreviated as S.P.

Profit or Gain: If the selling price of an article is more than the cost price, there is a gain or profit. Thus, Profit or Gain = S.P. - C.P.

Loss: If the cost price of an article is greater than the selling price, the seller suffers a loss. Thus, Loss = C.P. - S.P. respect to the cost price of the item.

Marked Price Formula (MPF)

Discount=Marked price- selling price

Discount percentage=(Discount/Marked price)*100

Example 1: (i) If C.P. = `235, S.P. = `240, then profit=?

(ii) If C.P. = `116, S.P. = `107, then loss =?

Solution: (i) Profit= S.P. - C.P. = 240 - 235 = rs.5.

(ii) Loss = C.P. - S.P. = 116 - 107 = rs.9.

Basic Formulae

01 Gain on rs100 is Gain per cent Gain

Gain%= $\frac{Gain \times 100}{C.P}$

Loss on rs100 Loss per cent

Loss%= $\frac{loss \times 100}{c.p}$

Another Example: A toy that cost 80 rupees is sold at a profit of 20 rupees. Find the percent or rate of

profit.

Solution: Gain/cost × 100 = % profit.

20/80 × 100 = 25%. - Answer

Example 2: The cost price of a shirt is rs.200 and selling price is rs. 250.Calculate the % ofprofit.

Solution: We have, C.P. = rs.200, S.P. = rs.250.

Profit= S.P. - C.P. = 250 - 200 = rs.50.

∴ Profit% $= \frac{profit \times 100}{C.P} = \frac{50 \times 100}{200}$ 25%

Example 3: Anu bought a necklace for rs.750 and sold it for rs.675. Find her percentage of loss.

Solution: Here, C.P. = rs.750, S.P. = rs.675

Loss = C.P. - S.P. = 750 - 675 = rs75.

$\therefore loss\% = \frac{loss \times 100}{C.P} = \frac{75 \times 100}{750} = 10\%$

02 When the selling price and gain%aregiven:

$$C.P = \left(\frac{100}{100 + Gain\%}\right) \times S.P$$

03 When the cost and gain per cent are given:

S.P=$\left(\frac{100+Gain\,\%}{100}\right)\times C.P$

Explanation

Since profit%=$\frac{profit\ \times 100}{C.P}$

$\therefore\frac{profit\ \%}{100}=\frac{S.P}{C.P}-1$

Or $\frac{S.P}{C.P}=1+\frac{profit\ \%}{100}$

$\therefore$ S.P. =$\left(\frac{100+profit\ \%}{100}\right)\ \times C.P$

and, C.P. =$\left(\frac{100}{100+profit\ \%}\right)\ \times S.P$

04 When the cost and loss per cent are given:

$$s.p=\left(\frac{100-loss\%}{100}\right)\times c.p$$

05 When the selling price and loss per cent are given:

C.P=$\left(\frac{100}{100-loss\,\%}\right)\times s.p$

Explanation

Since loss%=$\frac{loss\times 100}{C.P}$

=$\left[\frac{(C.P-S.P)\times 100}{C.P}\right]$

$\therefore\frac{loss\%}{100}=1-\frac{s.p}{c.p}$

Or $\frac{S.P}{C.P}=1-\frac{loss\%}{100}$

$\therefore$S.P=$\left(\frac{100-loss\,\%}{100}\right)\times C.P$

And, C.P=$\left(\frac{100}{100-loss\,\%}\right)\times S.P.$

Another Example: A damaged chair that cost Rs.110 was sold at a loss of 10%. Find the loss and the selling price.

Solution: Cost x percent loss = loss. 110 x 1/10 = 11, loss. Cost - loss = selling price. 110 - 11 = 99, selling price.

Example 4: Mr Sharma buys a cooler for rs.4500. For how much should he sell it to gain 8%?

Solution: We have, C.P. = rs.4500, gain%=8%

$\therefore$S.P=$\left(\frac{100+Gain\,\%}{100}\right)\times C.P$

=$\left(\frac{100+8}{100}\right)\times 4500$

=$\frac{108}{100}\times 4500=rs.4860$

Example 5: By selling a fridge for rs.7200, Pankaj loses10%. Find the cost price of the fridge.

Solution: We have, S.P. = rs.7200, loss% =10%

$\therefore$ C.P=$\left(\frac{100}{100-loss\,\%}\right)\times s.p$

=$\left(\frac{100}{100-10}\right)\times 7200$

$=\frac{100}{90} \times 7200 = rs. 8000$

To find the profit and the cost when the selling price and the percent profit are given, multiply the selling price by the percent profit and subtract the result from the selling price.

Another Example: A toy is sold for Rs. 6.00 at a profit of 25% of the selling price. Separate this selling price into cost and profit.

Answer :

Selling price x % profit = profit.

Selling price = profit + cost.

6.00 x .25 = 1.50, profit.

6.00 - 1.50 = 4.50, cost

Example 6: By selling a pen for rs.99, Mohan gains $12\frac{1}{2}$%. Find out cost price of the pen.

Solution: Here, S.P. = rs.99,gain% = 12

$\therefore C.P = \left(\frac{100}{100+GAIN\%}\right) \times s.p$

$=\left(\frac{100}{100+\frac{25}{2}}\right) \times 99$

$=\left(\frac{100\times 2}{225}\right) \times 99 = rs88$

5.1 Some Useful short cut methods

01 If a man buys x items for `y and sells z items for `w, then the gain or loss per cent made by him is

$$\left(\frac{xw}{zy} - 1\right) \times 100\%$$

Explanation S.P. of z iterms =rs.w

S.P. of x items = rs.$\frac{w}{z}x$

Net profit= $\frac{w}{z}x - y$

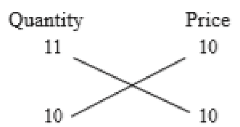
$\therefore$ %profit=$\frac{\frac{w}{z}x-y}{y} \times 100\%$

i.e,$\left(\frac{xw}{zy} - 1\right) \times 100\%$

which represents loss, if the result is negative.

Example 7: If 11 oranges are bought for `10 and sold at 10 for rs.11, what is the gain or loss %?

Solution:



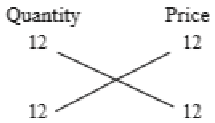
$\therefore$ %profit=$\left(\frac{xw}{zy} - 1\right) \times 100\%$

$=\left(\frac{11\times11}{10\times10}-1\right)\times100\%$

$=\frac{21}{100}\times100\%=21\%$

Example 8:A fruit seller buys apples at the rate of rs.12 per dozen and sells them at the rate of 15 for rs.12. Find out his percentage gain or loss.

Solution:



%gain or loss=$\left(\frac{xw}{zy}-1\right)\times100\%$

$=\left(\frac{12\times12}{15\times12}-1\right)\times100\%$

$$\left(\frac{144}{180}-1\right)\times100\%$$

$$\left(\frac{-36}{180}\times100\%\right)$$

Since the sign is -ve, there isa loss of 25%

02 If the cost price of m articles is equal to the selling price of n articles, then

%gain or loss=$\left(\frac{m-n}{n}\right)\times100$

[If m >n,it is % gain and, if m < n, it is % loss]

Explanation

Let, the C.P. of an article be rs.1.

∴ C.P. of m articles = `m × 1 = rs.m

S.P. of n articles = rs.m

∴ S.P. of an article =rs.$\frac{m}{n}$

∴ Profit on 1 article =rs$\left(\frac{m}{n}-1\right)$

i.e.,rs.$\left(\frac{m-n}{n}\right)$

∴%profit=$\left(\frac{m-n}{n}\right)\times100$i.e,$\left(\frac{m-n}{n}\right)\times100$

Example 9: A shopkeeper professes to sell his goods on cost price, but uses 800 gm, instead of 1 Kg. What is his gain %?

Solution: Here, cost price of 1000 gm is equal to selling price of 800 gm,

∴%gain=$\left(\frac{m-n}{n}\right)\times100$

$=\left(\frac{1000-800}{800}\right)\times100$

$=\frac{200}{800}\times100=25\%$

Example 10: If the selling price of 12 articles is equal to the cost price of 18 articles, what is the Profit %?

Solution: Here, m = 18, n = 12

∴ Profit $\% = \left(\frac{m-n}{n}\right) \times 100$

$= \left(\frac{1000-800}{800}\right) \times 100$

$= \frac{200}{800} \times 100 = 25\%$

03 If an article is sold at a price S.P.1., then % gain or % loss is x and if it is sold at a price S.P2., then % gain or % loss is y. If the cost price of the article is C.P., then

$$\frac{S.p}{100+x} = \frac{S.p_2}{100+y} = \frac{c.p}{100} = \frac{S.p_1 - s.p_2}{x-y}$$

where x or y is -ve, if it indicates a loss, otherwise it is +ve.

Example 11: By selling a radio for rs. 1536, Suresh lost 20%. What per cent shall he gain or lose by selling it for rs. 2000?

Solution: Here, $S.P_1$ = 1536, x = - 20

(-ve sign indicates loss)

$S.P_2$ = RS.2000, y = ?

Using the formula:

$$\frac{s.p_1}{100+x} = \frac{s.p_2}{100+y}$$

we get, $\frac{1536}{100-20} = \frac{2000}{100+y}$

$\Rightarrow 100+y = \frac{2000 \times 80}{1536} = 104\frac{1}{6}$

$\Rightarrow y = 4\frac{1}{6}\%$

Thus, Suresh has a gain of $4\frac{1}{6}\%$ by selling it for rs.2000

04 I f 'A' sells an article to 'B' at a gain/loss of m% and 'B' sells it to 'C' at a gain/loss of n%. If 'C' pays `z for it to 'B', then the cost price for 'A' is

$$\left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

Where m or n is -ve, of it indicates a loss, otherwise it is +ve.

Example 12: Mohit sells a bicycle to Rohit at a gain of 10% and Rohit again sells it to Jyoti at a profit of 5%. If Jyoti pays rs.462 to Rohit, what is the cost price of the bicycle for Mohit?

Solution: Here, m = 10, n = 5, z = rs.462.

Using the formula,

$$C.P = \left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

we get, C.P. for Mohit $=\left[\frac{100^2\times462}{(100+10)(100+5)}\right]$

=

$$\frac{462\times10000}{110\times105}=rs.400$$

Example 13: 'A' sells a DVD to 'B' at a gain of 17%and 'B' again sells it to 'C' at aloss of 25%. If 'C' pays `1053 to 'B', what is the cost price of the DVD to 'A'?

Solution: We have, m = 17, n = -25, z = rs.1053.

∴ Cost price of DVD to

$=\left[\frac{100^2z}{(100+m)(100+n)}\right]$

$=\frac{100\times100\times1053}{(100+17)(100-25)}$

$=\frac{100\times100\times1053}{117\times75}=rs.1200$

05 If 'A' sells an article to 'B' at a gain/loss of m%, and 'B' sells it to 'C' at a gain/loss of n%, then the resultantProfit/loss percent is given by

$\left(m+n+\frac{mn}{100}\right)$ (1)

Where m or n is -ve, if it indicates a loss, otherwise it is +ve.

Example 14: 'A' sells a horse to 'B' at a profit of 5% and 'B' sells it: to 'C' at a profit of 10%. Find out the resultantprofit percent.

Solution: We have, m = 5 and n = 10.

∴ Resultant profit %= $\left(m+n+\frac{mn}{100}\right)$

$$\left(5+10+\frac{5\times10}{100}\right)$$

$=\frac{31}{2}\%\ or\ 15\frac{1}{2}$

Example 15: Manoj sells a shirt to Yogesh at a profit of 15%, and Yogesh sells it to Suresh at a loss of 10%. Find the resultantprofit or loss.

Solution: Here, m = 15, n = -10

∴ Resultant profit/loss % $=\left(m+n+\frac{mn}{100}\right)$

$=\left(15-10+\frac{15\times-10}{100}\right)$ $=\left(15-10-\frac{150}{100}\right)$

=7/2% or $3\frac{1}{2}$%

which represents profit as the sign is +ve.

06 When two different articles are sold at the same selling price, getting gain/loss of x% on the first and gain/loss of y% on the second, then the overall % gain or % loss in the transaction is given by

$$\left[\frac{100(x+y)+2xy}{(100+x)+(100+y)}\right]\%$$

The above expression represents overall gain or loss according as its sign is +ve or -ve.

07 When two different articles are sold at the same selling price getting a gain of x% on the first and loss of x% on the second, then the overall % loss in the transaction is given by

$$\left(\frac{x}{10}\right)^2\%$$

Note that in such questions, there is always a loss.

Explanation

Let, each article be sold at rs.z.

Since gain/loss of x% is made on the first, cost price of the first article

=rs.z$\left(\frac{100}{100+x}\right)$

Also, gain/loss of y% is made on the second. Therefore, cost price of the second article

=rs. z$\left(\frac{100}{100+y}\right)$

∴ Total C.P=z$\left(\frac{100}{100+x}\right)+z\left(\frac{100}{100+y}\right)$

=z$\left[\frac{100(100+y)+100(100+x)}{(100+x)(100+y)}\right]$

Total S.P. = 2z.

∴ Overall % gain or loss =$\frac{S.P-C.P}{C.P}\times 100$

=$\frac{2z-\frac{100z\{100+x+100+y\}}{(100+x)(100+y)}}{\frac{100z\{100+x+100+y\}}{(100+x)(100+y)}}\times 100$

=$\frac{2(100+x)(100+y)-100(200+x+y)}{100(200+x+y)}\times 10$

=$\frac{100x+100y+2xy}{(100+x)(100+y)\%}$

=$\left[\frac{100(x+y)+2xy}{(100+x)+(100+y)}\right]\%$

Example 16: Mahesh sold two scooters, each for `24000. If he makes 20% profit on the first and 15% loss on the second, what is his gain or loss per cent in the transactions?

Solution: Here, x = 20 and y = -15.

∴ Over all gain/loss %

=$\left[\frac{100(x+y)+2xy}{(100+x)+(100+y)}\right]\%$

=$\left[\frac{100(20-15)+2\times 20\times -15}{(100+20)+(100-15)}\right]\%$

=-$\frac{100}{205}\%=-\frac{20}{41}\%$

which represents loss, being a -ve expression.

Example 17: Rajesh sold two horses for `990 each; gaining 10% on the one and losing 10% on the other. Find out his total gain or loss per cent.

Solution: Here, x = 10.

∴ Overall loss % $=\left(\frac{x}{10}\right)^2 \% = \left(\frac{10}{10}\right)^2 \% = 1\%$

08 A merchant uses faulty measure and sells his goods at gain/loss of x%. The overall % gain/loss(g) is given by

$$\frac{100+g}{100+x} = \frac{Trye\ measure}{Faulty\ measure}$$

09 A merchant uses y% less weight/length and sells his goods at gain/loss of x%.The overall %gain/ loss is given by

$$\left[\left(\frac{y+x}{100-y}\right) \times 100\right]\%$$

5.2 False weight

Example 18: A dishonest shopkeeper professes to sell cloth at the cost price, but he uses faulty meter rod. His meter rod measures 95 cm only. Find his gain per cent.

Solution: Here, true measure = 100 cm.

False measure = 95 cm.

Since the shopkeeper sells the cloth at cost price, ∴ x = 0. ∴Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{True\ measure}{False\ measure}$$

$\Rightarrow \frac{100+g}{100} = \frac{100}{95} \Rightarrow 100+g=\frac{100\times100}{95}$

$\Rightarrow g=\frac{10000}{95} - 100 = 5\frac{5}{19}\%$

Example 19: A dishonest shopkeeper professes to sell his goods at cost price, but he uses a weight of 800 g for the Kg weight. Find out his gain per cent.

Solution: True measure = 1000 g. False measure = 800 g. Also, x = 0.

∴ Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{True\ measure}{False\ measure}$$

$\Rightarrow \frac{100+g}{100} = \frac{1000}{800} \Rightarrow 100+g=\frac{1000\times100}{800}$

$\Rightarrow g=\frac{1000}{8} - 100 = 25\%$

Example 20: A shopkeeper sells goods at 44% loss on cost price, but uses 30% less weight. What is his percentage profit or loss?

Solution: Here, x = -44 and y = 30.

∴ Overall gain/loss% $=\left(\frac{y+x}{100-y}\right)\times 100\%$

$=\left(\frac{30-44}{100-30}\times 100\right)\%$

$=\left(\frac{-14}{70}\times 100\right)\% = -20\%$

which represents loss being a negative expression.

10 A person buys two items for `A and sells one at a loss of l% and the other at a gain of g%. If each item was sold at the same price, then

(a) The cost price of the item sold at loss

$=\frac{A(100+\%gain)}{(100-\%loss)+(100+\%gain)}$

(b) The cost price of the item sold at gain

$$\frac{A(100+\%gain)}{(100-\%loss)+(100+\%gain)}$$

Example 21: Ramesh buys two books for rs.410. He sells one at a loss of 20% and the other at a gain of 25%. If both the books are sold at the same price, find out the cost price of two books.

Solution: Cost price of the book sold at a loss of 20%

$=\frac{410(100+25)}{(100-20)+(100+25)}$

$=\frac{410\times 125}{80+125} = rs.250$

Cost price of the book sold at a profit of 25%

$=\frac{410(100-20)}{(100-20)+(100+25)} = \frac{410\times 80}{80+125}$=rs.160.

5.3 Successive Discount and marked price

11 If two successive discounts on an article are m% and n% respectively, then a single discount equivalent to the two successive discounts will be:

$$\left(m+n-\frac{mn}{100}\right)\%$$

Explanation

Let, the marked price of the article be rs.100.

∴ S.P. after the first discount = rs. (100 - m) and discount at n% on rs(100-m)=rs.$\frac{(100-m)\times n}{100}$

∴ Single equivalent discount

$=\left[m+\frac{(100-m)\times n}{100}\right]\%$

$=\left(\frac{100m+100n-mn}{100}\right)\%$

$=\left(m+n-\frac{mn}{100}\right)\%$

12 If three successive discounts on an article are l%, m% and n% respectively, then a single discount equivalent to the three successive discounts will be

$$\left[l+m+n-\frac{(lm+ln+mn)}{100}+\frac{lmn}{100^2}\right]\%$$

Explanation

Let, the marked price of the article be rs.100.

∴ S.P. after the first discount = rs.(100 - l).

Second discount at m% on rs.(100 - l)

=rs.$\left(\frac{(100-l)\times m}{100}\right)$

∴ S.P. after second the discount

=rs.(100-l)-$\left(\frac{(100-l)\times m}{100}\right)$

=rs.$\frac{100(100-l)-(100-l)m}{100}$

=rs.$\frac{(100-l)(100-m)}{100}$

Third discount at n% on rs.$\frac{(100-l)(100-m)}{100}$

=rs.$\frac{(100-l)(100-m)}{100\times 100}$

∴ S.P. after the third discount

=rs. $\frac{(100-l)(100-m)}{100} - \frac{(100-l)(100-m)n}{100\times 100}$

=rs.$\frac{(100-l)(100-m)(100-n)}{100\times 100}$

=$\left(l + m + n - \frac{(lm+ln+mn)}{100} + \frac{lmn}{(100)^2}\right)$

∴ Single equivalent discount

=$\left(l + m + n - \frac{(lm+ln+mn)}{100} + \frac{lmn}{(100)^2}\right)$ %

Example 22: Find a single discount equivalent to two successive discounts of 10% and 20%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 - \frac{10 \times 20}{100}\right)\% i.e..28\%$$

Example 23: Find out a single discount equivalent to three successive discounts of 10%, 20% and 30%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 + 30 - \frac{(10 \times 20 + 10 \times 30 + 20 \times 30)}{100} + \frac{10 \times 20 \times 30}{100^2}\right)\%$$

i.e$\left(60 - 11 + \frac{6}{10}\right)\% = \frac{496}{10}\% or \ 49.6\%$

Example 24: Two shopkeepers sell machines at the same list price. The first allows two successive discounts of 30% and 16% and the second 20% and 26%.Which discount series is more advantageous to the buyers?

Solution: A single discount equivalent to the two successive discounts of 30% and 16% is

$$\left(30 + 16 - \frac{301 \times 16}{100}\right)\%$$

Or, $\left(46 - \frac{26}{5}\right)\%\ or\ 41\frac{1}{5}\%$

Also, a single discount equivalent to the two successive discounts of 20% and 26% is

$$\left(20 + 26 - \frac{20 \times 26}{100}\right)\%$$

Or, $\left(46 - \frac{26}{5}\right)\%\ or\ 40\frac{4}{5}\%$

Clearly, the discount series being offered by the first shopkeeper is more advantageous to the buyers.

13 A shopkeeper sells an item at rs.z after offering a discount of d% on labelled price. Had he not offered the discount, he would have earned a profit of p% on the cost price.

The cost price of each item is given by

$$C.P = \left[\frac{100^2 Z}{(100 - d)(100 + p)}\right]$$

Example 25:A shopkeeper sold sarees at `266 each after giving 5% discount on labelled price. Had he not given the discount, he would have earned a Profit of 12% on the cost price. What was the cost price of each saree?

Solution: We have, labelled price z = rs.266, discount d = 5% and profit p = 12% Using the formula

$$C.P = \left[\frac{100^2 Z}{(100 - d)(100 + p)}\right]$$

we get the cost price of each saree

$$\left[\frac{100 \times 100 \times 266}{(100 - 5)(100 + 12)}\right]$$

$=\frac{100 \times 100 \times 266}{(100-5)(100+12)}$

Summary

The key concepts learned from this unit are: -

- We have learnt how to calculate Cost Price, Selling Price and Profit or Gain.
- We have learnt how to calculate successive discount and marked price

Keywords

- Cost Price
- Selling Price
- Profit
- Gain.

- discount price
- Marked price

Self Assessment

1. A man buys an article for rs.27.50 and sells it for rs.28.50. His gain % is?

A. 1.
B. 2.
C. 3.
D. 4.

2. If the a radio is sold for rs 490 and sold for rs 465.50. The loss% is?

A. 2
B. 3
C. 4
D. 5

3. S.P when CP=56.25, gain=20% is ?

A. 67
B. 68
C. 67.5
D. 68

4. CP when SP = rs 40.60, gain=16% is?

A. 30
B. 35
C. 40
D. 45

5. A person incures loss for by selling a watch for rs1140 at what price should the watch be sold to earn a 5% profit ?

A. 1260
B. 3300
C. 450
D. 50

6. By selling 33 metres of cloth , one gains the selling price of 11 metres . Then the gain percent is?

A. 50
B. 55
C. 60
D. 65

7. If the cost price is 96% of sp then what is the profit %?

A. 2.32
B. 3.67
C. 4.17

D. 5.98

8. A man brought toffees at for a rupee. How many for a rupee must he sell to gain 50%?

A. 1
B. 2
C. 4
D. 5

9. A grocer purchased 80 kg of sugar at Rs.13.50 per kg and mixed it with 120kg sugar at Rs.16per kg. At what rate should he sell the mixer to gain 16%?

A. 30.2
B. 20.3
C. 17.4
D. 51.8

10. A dishonest dealer professes to sell his goods at cost price but uses a weight of 960 gms for a kg weight . Find his gain percent.

(a) 3.2
(b) 4.1
(c) 1.8
(d) 5.8

11. Monika purchased a pressure cooker at 9/10th of its selling price and sold it at 8% more than its S.P .Then her gain percent is?

A. 20
B. 12
C. 17
D. 4

12. If the manufacturer gains 10%,the wholesale dealer 15% and the retailer 25%,then find the cost of production of a ,the retail price of which is Rs.1265?.

(a) 900
(b) 800
(c) 200
(d) 300

13. A An article is sold at certain price. By selling it at 2/3 of its price one losses 10%,find the gain at original price?

A. 30
B. 20
C. 35
D. 51

14. A the single discount equivalent to a series discount of 20% ,10% and 5% is ?

A. 12
B. 76.2
C. 68.4

D. 55

15. A retailer buys 40 pens at the market price of 36 pens from a wholesaler, if he sells these pens giving a discount of 1%, what is the profit %?

A. 30
B. 20
C. 15
D. 10

Answers for Self Assessment

1.	D	2.	D	3.	C	4.	B	5.	A
6.	B	7.	C	8.	B	9.	C	10.	B
11.	A	12.	B	13.	C	14.	C	15.	D

Review Questions

1. At what % above C.P must an article be marked so as to gain 33% after allowing a customer a discount of 5%?
2. An uneducated retailer marks all its goods at 50% above the cost price and thinking that he will still make 25% profit, offers a discount of 25% on the market price. what is the actual profit on the sales?
3. A man bought a horse and a car riage for Rs 3000.he sold the horse at a gain of 20% and the carriage at a loss of 10%, thereby gaining 2% on the whole. find the cost of the horse.
4. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is?
5. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is?
6. If selling price is doubled, the profit triples. Find the profit percent.
7. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
8. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?
9. The cost price of three varieties of oranges namely A, B and C is Rs 20/kg, Rs 40/kg and Rs 50/kg. Find the selling price of one kg of orange in which these three varieties of oranges are mixed in the ratio of 2 : 3 : 5 such that there is a net profit of 20%?
10. A sweet seller sells 3/5th part of sweets at a profit of 10% and remaining at a loss of 5%. If the total profit is Rs 1500, then what is the total cost price of sweets?

Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing

3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle